

Automating Formulary Exception Adjudication

1 Overview

DaVita's dialysis operations are reimbursed by Medicare under a flat "bundled" rate per treatment. This bundle obligates DaVita to provide all necessary medications and supplies, creating a direct financial and clinical incentive to optimize drug selection. DaVita maintains a **Formulary** — a curated list of preferred drugs. When a physician prescribes something not on this list, a **Formulary Exception (FE)** must be requested and clinically justified before it can be approved.

Today, this FE adjudication process is entirely manual — 12 coordinators processing ~600 cases per day across 7 disconnected systems using 100+ clinical decision trees stored in PowerPoint files. This proposal presents a solution to digitize and automate that workflow, built on the **Clinical Golden Set** from the CDI Auditing project in BigQuery, supplemented by insurance and pharmacy operations data from HealthDyne and ROPS Nautilus. It directly supports DaVita's **Operations Excellence** and **Measurable Impact through Technology** ambitions and addresses three goals identified by PharmaOps: maintainable decision-tree management, automated approvals with CPM oversight, and a streamlined coordinator experience.

2 The Problem: 7 Systems, 100 Decision Trees, Manual Handoffs

A single FE case forces a **CPM Coordinator** through a gauntlet of disconnected systems. A **Field Teammate** submits a request in Pharos — often **multiple times**, restarting when data is incomplete. The coordinator then hunts through CWOW (allergies), Tableau (labs), HealthDyne (PA status), ROPS Nautilus (insurance auth), and OneDrive (decision trees). Missing data triggers an **Outlook email** to the clinic, stalling the case **1–7 days**. Once assembled, the coordinator **mentally walks a PPT/Visio decision tree**, renders a verdict, then **manually updates 3+ downstream systems** — iProc, WellDyne, PAHub, and potentially SHAPE protocols.

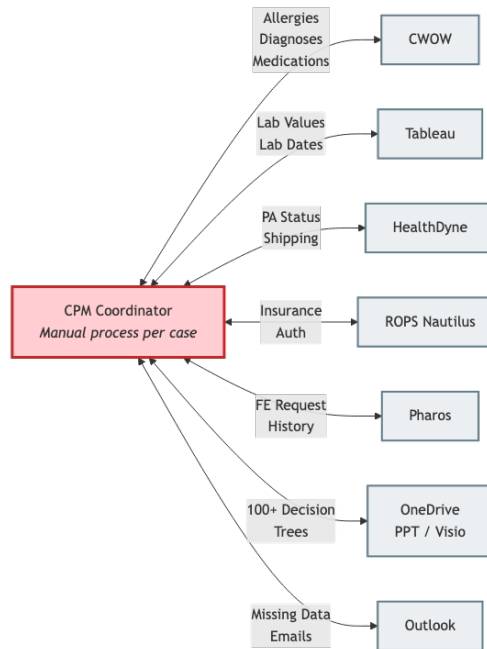


Figure 1 — Current state: CPM Coordinator as central hub across 7 systems

Metric	Current State
Daily FE volume	~600 cases
Coordinator headcount	12 CPMs
Throughput ceiling	~50 cases / coordinator / day
Turnaround time	1–7 days
Systems touched per case	7
Decision trees (PPT/Visio)	100+ — updated weekly, growing portfolio

This workflow is **headcount-bound** (more cases = hire more people), **error-prone** (expired lab data slips through manual date checks — **compliance risk** for CMS-regulated decisions), and **inconsistent** (two coordinators can reach different verdicts on the same case). Every FE is also a **cost decision** — each approved non-formulary drug has direct revenue impact on bundled reimbursement. There is no lever to pull except adding headcount.

3 The Solution: Consolidate the Data, Automate the Logic

The problem has two parts. The solution addresses each one directly.

3.1 The data problem — 4 lookups become 1 query, 2 systems disappear

Of the 7 systems touched today, patient clinical and treatment data (CWOW, Tableau) is **already in BigQuery** — the same **Clinical Golden Set** built through the CDI Auditing project. That foundation means the hardest data problem is already solved. The only remaining gap is **insurance and pharmacy operations** from HealthDyne and ROPS Nautilus — PA status, coverage eligibility, and shipping — which requires new ingestion into BigQuery. PA status changes infrequently (approvals last months), so scheduled batch ingestion is likely sufficient.

System	Data	Today	TO-BE
CWOW	Allergies, diagnoses, medications	Manual lookup	Already in BigQuery — CDI Clinical Golden Set
Tableau source	Lab values (K+, Hgb), drawn dates	Manual lookup	Already in BigQuery — CDI Clinical Golden Set
HealthDyne	PA status, shipping, insurance	Manual lookup	New ingestion needed — batch ETL into BigQuery
ROPS Nautilus	Insurance authorization queue	Manual lookup	New ingestion needed — batch ETL into BigQuery
OneDrive	100+ PPT/Visio decision trees	Manual reference	Eliminated — trees become executable rules
Outlook	"More info needed" emails to clinics	Manual email	Eliminated — pre-fetched data removes the scenario
Pharos	FE request submission & verdicts	Manual entry	No change — request trigger and verdict display

When an FE request arrives from Pharos, the Rules Engine queries BigQuery for that patient's combined profile — clinical data from the Golden Set, insurance data from the new pipeline — and returns everything in a single response. This comprehensive record is the single input the decision tree evaluates against.

3.2 The decision problem — PPT trees become executable rules

A **Rules Engine** encodes the exact same clinical decision trees coordinators use today as executable, deterministic logic — digitized, versioned, testable, and auditable, running the same clinical logic faster and with zero ambiguity. Each rule set handles full product complexity — new requests vs. renewals, modality-specific paths (ICHHD vs. PD/HHD), and product-specific qualifying conditions. Decision trees are updated weekly today; in the Rules Engine, updates follow the same cadence — modify the rule, validate against test cases, deploy — versioned and auditable, no full system redeployment required. This is the enterprise destination; Section 3.3 explains how we get there in two stages.

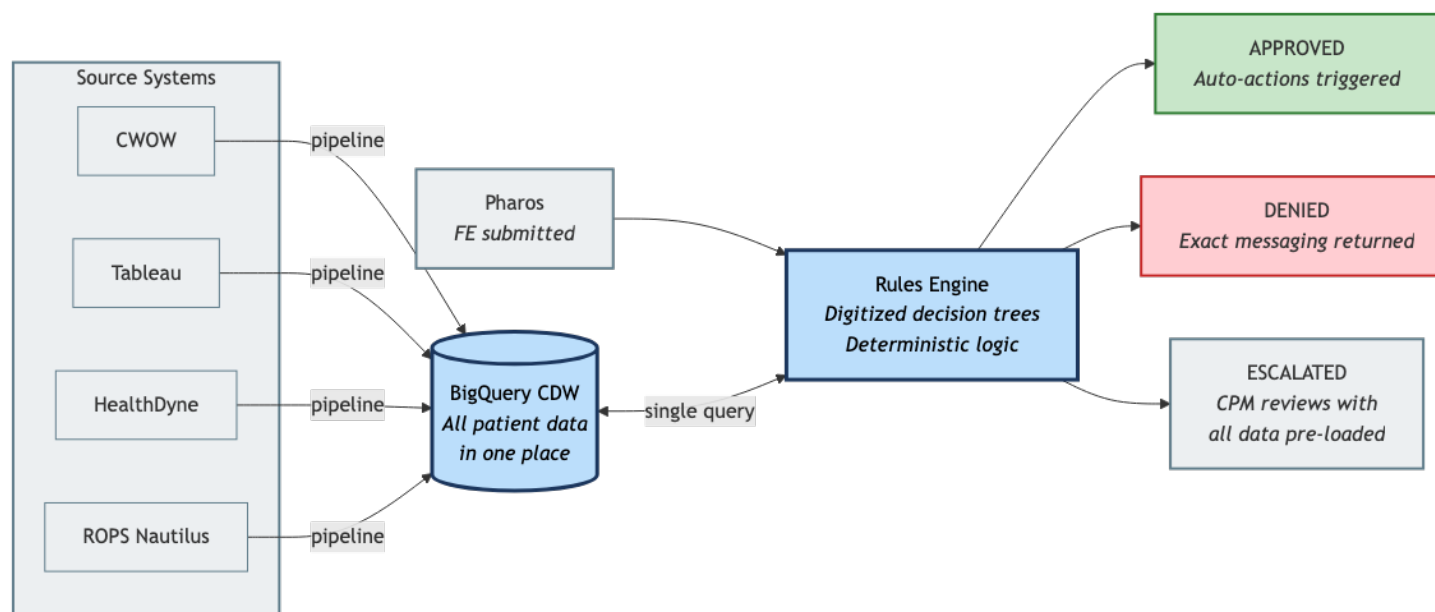


Figure 2 — Target state: Source systems → BigQuery → Rules Engine → outcomes

3.3 From PowerPoint to executable logic

Today the portfolio spans 300+ slides of branching logic — all maintained in PowerPoint, updated weekly. Some trees are simple (Dialysate: 3 decision points); the most complex — **Epogen MDV** — chains 6 steps across clinical eligibility gates, modality branching, special-condition checks, dosing logic, and approval-duration rules, each referencing a different source system. A coordinator walks this **mentally**, 50 times a day. The question is: *how does this become executable automation without a months-long extraction effort?*

The answer is **two consumption modes**, matched to PoC phases. Each encoded tree is a **Skill** — a self-contained unit of clinical logic the AI can execute:

Image mode (Pilot) — Today's best multimodal AI models can read visual decision trees directly — following branching logic, evaluating thresholds, and applying conditional rules from a flowchart image. **No extraction pipeline needed: the AI reads the same PowerPoint the coordinator uses today.** The model evaluates the patient's data against the flowchart and delivers a structured recommendation; the CPM verifies. **When the clinical team updates a tree — which happens weekly — the AI simply reads the new version.** A new product onboards in hours, and the portfolio grows without engineering effort.

Rules mode (Enterprise) — Each branch is formally encoded as deterministic if/then logic — testable, versioned, and fully auditable. No AI interpretation: pure rule execution. Clear cases auto-adjudicate; edge cases escalate to a clinician. **When a tree is updated, the AI drafts updated rules from the new PowerPoint — a human validates against test cases before deployment.** Maintenance is AI-assisted, not eliminated; the work shifts from manual re-encoding to reviewing an AI-generated diff.

Why two modes? Image mode proves the logic works in days with zero engineering investment. Once concordance is validated, the proven logic is hardened into formal rules for autonomous execution. The pilot earns the right to the enterprise build — and the clinical validation carries over directly. In practice, the two modes coexist: formal rules for the stable core, image mode as a rapid onboarding path for new products not yet encoded.

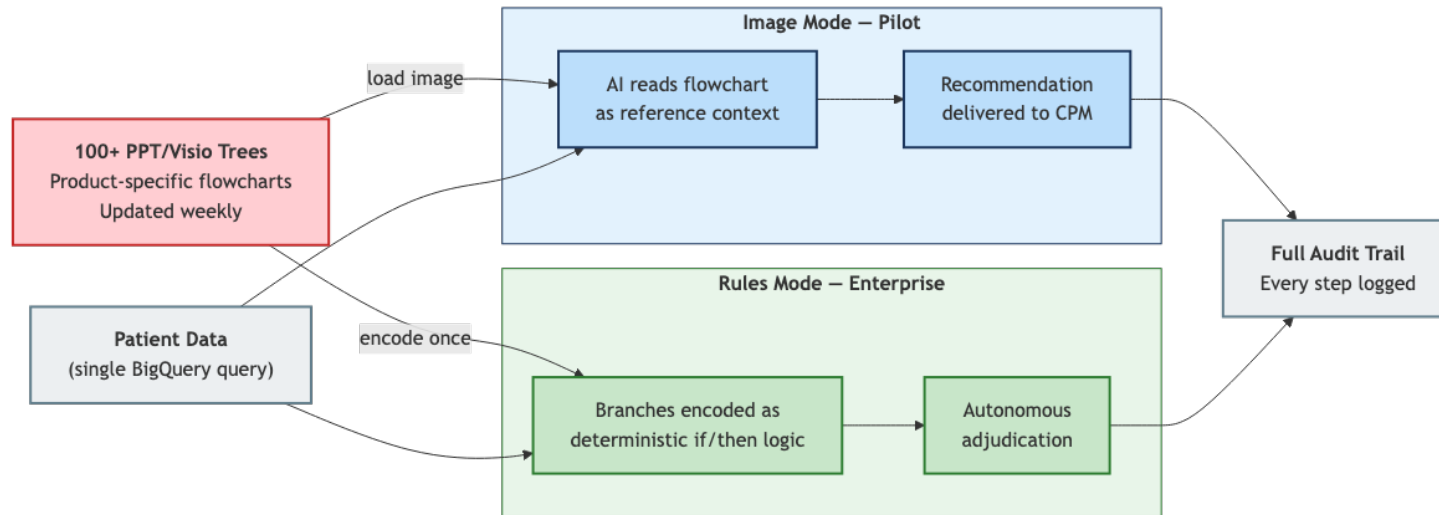


Figure 3 — Same decision tree, two consumption modes: pilot proves the logic, enterprise hardens it

3.4 How a case flows through the engine

A concrete example — Dialysate 1K Bath request for an ICHD patient. The logic is identical in both modes; only the execution mechanism differs:

REQUEST: Dialysate 1K Bath | Patient: ICHD modality | K+ = 4.2, drawn 2026-03-10

ENGINE: Load Dialysate rule set → check qualifying conditions → check modality (ICHD)
→ check lab recency: 18 days since drawn, threshold is 14 → FAIL

VERDICT: DENIED — "Lab results older than 14 days. Please obtain updated labs and resubmit."

TIME: milliseconds | AUDIT: every step logged

Same decision a coordinator would make — in milliseconds, with zero ambiguity on the date math, and a complete audit record. Every case produces one of three outcomes:

Outcome	What Happens	Human Involvement
APPROVED	Auto-triggers downstream actions: iProc ordering (M10/M20), WellDyne Rx, PAHub auth, and SHAPE protocol activations where applicable	None
DENIED	Returns exact, clinically-validated denial messaging to Pharos with specific resubmission guidance for the Field Teammate	None
ESCALATED	Routes to CPM Dashboard — a single screen with all patient data pre-loaded, the engine's evaluation trace, and action buttons	CPM reviews and decides — data gathering already done

CPMs do not go away. The engine escalates anything it cannot adjudicate with certainty — all clinical denials are made by a clinician. Coordinators shift from data-entry work to clinical judgment on genuinely complex cases. The CPM Dashboard is an operational layer delivered during Phase 2, addressing existing Pharos pain points without modifying Pharos itself.

4 How We Prove It: PoC Approach

One question before we invest: *can the clinical decision trees be reliably digitized and automated?* Three phases answer it — each earns the right to the next.

Phase 1 — Validate the Logic (Image Mode)

We start with 2–3 simple products (Dialysate, Sorbaview, Extraneal). Each decision tree is loaded as a visual reference — the AI reads the PPT/Visio flowchart directly, walks the branching logic against structured patient data, and produces a verdict. We replay historical cases where we already know the correct decision. The question is binary: *does the AI match CPM decisions?* No formal encoding, no live systems affected, no patient impact — just logic validation in days.

Phase 2 — Pilot with Live Users (Image Mode)

Once the logic validates, we put it in front of real CPMs. An AI assistant runs the same image-mode approach in the background — pulling the patient profile from BigQuery, reading the decision tree, and delivering a structured recommendation with full audit trail. CPMs still own the final

decision; the assistant handles data gathering and rule evaluation. Anthropic's **Claude Co-Worker** enables this — the same approach built in Phase 1 runs directly inside it with zero re-engineering, so deployment takes days rather than a full infrastructure build.

Phase 3 — Measure & Build the Enterprise Case (Rules Mode)

The pilot runs alongside CPMs in shadow mode. We compare its recommendations against CPM decisions over a sustained period and collect the data that matters: concordance rate, processing time per case, CPM time savings, and coordinator throughput ratio (currently 1:50 cases/day). Pioneer time studies of CPM workflow are already in progress — pilot results will complement that baseline. Key gate: **≥95% concordance** with CPM decisions before scaling.

This phase produces the evidence to transition from image mode to **rules mode** — a **full GCP-native production system** with each decision tree formally encoded as deterministic logic on BigQuery, Cloud Run, and Vertex AI with CI/CD, monitoring, and enterprise governance. Instead of committing to a multi-month GCP build on projected value, leadership decides based on *measured* concordance and throughput from live cases. The enterprise architecture is the destination — the pilot is how we prove the business case to get there.

Phase	Goal	Key Activity
Now	Groundwork complete	Decision trees mapped. System dependencies documented. Architecture designed.
1 — Validate <i>Image mode</i>	Logic works	Load decision trees as images. AI reads flowcharts, replays historical cases. Measure concordance.
2 — Pilot <i>Image mode</i>	Live user feedback	AI assistant delivers recommendations to CPMs in real time. No infrastructure build required.
3 — Measure <i>Rules mode</i>	Enterprise case	Concordance, time-per-case, CPM throughput gains → evidence for formal GCP rules-mode production build.

If any phase fails to meet its goal, we stop at minimal cost.

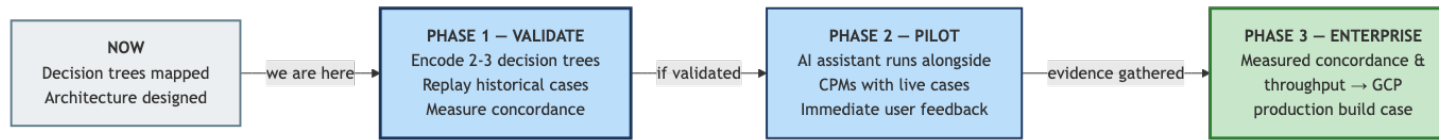


Figure 4 — PoC progression: validate, pilot, measure

5 The Tools

Tool	Role
Claude	AI model — reads decision trees, evaluates patient data, produces verdicts with audit trails
Claude Code	Rapid Skill development — encode and validate a decision tree in hours
Claude Co-Worker	Autonomous agent — runs Skills in background, delivers recommendations to CPMs
BigQuery + Vertex AI	Clinical Data Warehouse + LLM inference — already deployed; single patient data layer
Python	Concordance analysis and test harness for validating against historical outcomes

6 Conclusion

The solution consolidates 4 manual system lookups into a single patient query, eliminates 2 systems entirely (decision tree documents and email chase), and replaces 100+ PowerPoint decision trees with a deterministic, auditable rules engine — while keeping CPM coordinators in the loop for complex clinical judgment. Every rule evaluation is logged, every denial uses clinically-validated messaging, and every ambiguous case escalates to a human clinician.

The path forward is three phases: **validate** the logic in image mode against historical cases, **pilot** with live CPMs to collect real concordance and throughput data, then use that evidence to greenlight the transition to **rules mode** — a full GCP-native production system. The immediate next step is Phase 1: load 2–3 decision trees and measure whether the AI reaches the same verdicts as coordinators.